Q1.

A curve is defined parametrically by

$$x = t^3 + 5, \qquad y = 6t^2 - 1$$

The arc length between the points where $t = 0$ and $t = 3$ on the curve is s.

(a) Show that $s = \int_0^3 3t\sqrt{t^2 + A}\,\mathrm{d}t$, stating the value of the constant A.

(4)

(b) Hence show that $s = 61$.

(4)

(Total 8 marks)

Q2.

A curve has cartesian equation

$$y = \frac{1}{2}\ln(\tanh x)$$

(a) Show that

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{\sinh 2x}$$

(4)

(b) The points A and B on the curve have x-coordinates ln 2 and ln 4 respectively. Find the arc length AB, giving your answer in the form $p \ln q$, where p and q are rational numbers.

(8)

(Total 12 marks)

Q3.

(a) Given that

$$x = \ln(\sec t + \tan t) - \sin t$$

show that

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \sin t \tan t$$

(4)

(b) A curve is given parametrically by the equations

$$x = \ln(\sec t + \tan t) - \sin t, \quad y = \cos t$$

The length of the arc of the curve between the points where $t = 0$ and $t = \frac{\pi}{3}$ is denoted by s.

Show that $s = \ln p$, where p is an integer.

(6)
(Total 10 marks)

Q4.

(a) Using the identities

$$\cosh^2 t - \sinh^2 t = 1,\ \tanh t = \frac{\sinh t}{\cosh t} \text{ and } \operatorname{sech} t = \frac{1}{\cosh t}$$

show that:

(i) $\tanh^2 t + \operatorname{sech}^2 t = 1$;

(2)

(ii) $\frac{d}{dt}(\tanh t) = \operatorname{sech}^2 t$;

(3)

(iii) $\frac{d}{dt}(\operatorname{sech} t) = -\operatorname{sech} t \tanh t$.

(3)

(b) A curve C is given parametrically by

$$x = \operatorname{sech} t,\ y = 4 - \tanh t$$

(i) Show that the arc length, s, of C between the points where $t = 0$ and $t = \frac{1}{2}\ln 3$ is given by

$$s = \int_0^{\frac{1}{2}\ln 3} \operatorname{sech} t \, dt$$

(4)

(ii) Using the substitution $u = e^t$, find the exact value of s.

(6)
(Total 18 marks)

Q5.

(a) Show that

$$\frac{d}{dx}\left(\cosh^{-1}\frac{1}{x}\right) = \frac{-1}{x\sqrt{1-x^2}}$$

(3)

(b) A curve has equation

$$y = \sqrt{1 - x^2} - \cosh^{-1}\frac{1}{x} \qquad (0 < x < 1)$$

Show that:

(i) $\dfrac{dy}{dx} = \dfrac{\sqrt{1 - x^2}}{x}$;

(4)

(ii) the length of the arc of the curve from the point where $x = \frac{1}{4}$ to the point where $x = \frac{3}{4}$ is ln 3.

(5)

(Total 12 marks)

Q6.

The diagram shows a curve which starts from the point A with coordinates (0, 2). The curve is such that, at every point P on the curve,

$$\frac{dx}{dy} = \frac{1}{2}s$$

where s is the length of the arc AP.

(a) (i) Show that

$$\frac{ds}{dx} = \frac{1}{2}\sqrt{4 + s^2}$$

(3)

(ii) Hence show that

$$s = 2\sinh\frac{x}{2}$$

(4)

(iii) Hence find the cartesian equation of the curve.

(3)

(b) Show that

$$y^2 = 4 + s^2$$

(2)
(Total 12 marks)

Q7.

(a) Given that $y = \ln \tanh \frac{x}{2}$, where $x > 0$, show that

$$\frac{dy}{dx} = \operatorname{cosech} x$$

(6)

(b) A curve has equation $y = \ln \tanh \frac{x}{2}$, where $x > 0$. The length of the arc of the curve between the points where $x = 1$ and $x = 2$ is denoted by s.

(i) Show that

$$s = \int_1^2 \coth x \, dx$$

(2)

(ii) Hence show that $s = \ln(2 \cosh 1)$.

(4)
(Total 12 marks)

Q8.

(a) Given that $y = \operatorname{sech} t$, show that:

(i) $\frac{dy}{dt} = -\operatorname{sech} t \tanh t$

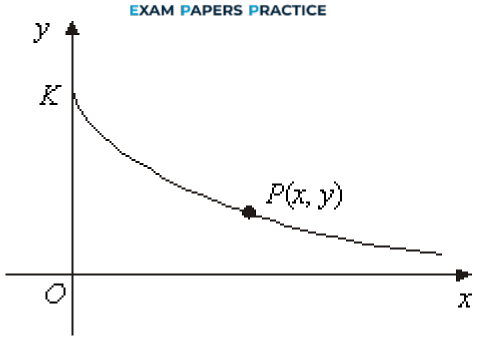
(3)

(ii) $\left(\frac{dy}{dt}\right)^2 = \operatorname{sech}^2 t - \operatorname{sech}^4 t$.

(2)

(b) The diagram shows a sketch of part of the curve given parametrically by

$$x = t - \tanh t \qquad y = \operatorname{sech} t$$



The curve meets the y-axis at the point K, and $P(x, y)$ is a general point on the curve. The arc length KP is denoted by s. Show that:

(i) $$\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \tanh^2 t;$$

(4)

(ii) $s = \ln \cosh t$;

(3)

(iii) $y = e^{-s}$.

(2)

(c) The arc KP is rotated through 2π radians about the x-axis. Show that the surface area generated is $2\pi\,(1 - e^{-s})$

(4)

(Total 18 marks)

Q9.

A curve has equation $y = 4\sqrt{x}$.

(a) Show that the length of arc s of the curve between the points where $x = 0$ and $x = 1$ is given by

$$s = \int_0^1 \sqrt{\frac{x+4}{x}}\,dx$$

(4)

(b) (i) Use the substitution $x = 4\sinh^2\theta$ to show that

$$\int \sqrt{\frac{x+4}{x}}\,dx = \int 8\cosh^2\theta\,d\theta$$

(5)

(ii) Hence show that $s = 4\sinh^{-1} 0.5 + \sqrt{5}$

(6)

(Total 15 marks)

Q10.

(a) Use the definitions

$\sinh\theta = \frac{1}{2}(e^{\theta} - e^{-\theta})$ and $\cosh\theta = \frac{1}{2}(e^{\theta} + e^{-\theta})$

to show that:

(i) $2\sinh\theta\cosh\theta = \sinh 2\theta$;

(2)

(ii) $\cosh^2\theta + \sinh^2\theta = \cosh 2\theta$.

(3)

(b) A curve is given parametrically by

$x = \cosh^3\theta$, $y = \sinh^3\theta$

(i) Show that

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = \frac{9}{4}\sinh^2 2\theta\cosh 2\theta$$

(6)

(ii) Show that the length of the arc of the curve from the point where $\theta = 0$ to the point where $\theta = 1$ is

$$\frac{1}{2}\left[(\cosh 2)^{\frac{3}{2}} - 1\right]$$

(6)

(Total 17 marks)